

Paper OOL-5: Oscillating Constraints and the Polymerization Ratchet: Environmental Cycling as Chemical Memory

Steve Bico Mujjabi, MD

Independent Researcher

ORCID: 0009-0001-0556-5516

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Abstract

This manuscript addresses the water paradox in origin-of-life chemistry: aqueous environments enable transport and encounter, while condensation chemistry is opposed by hydrolysis in bulk water. CDFD reframes the problem as a temporal constraint problem. Wet-dry, freeze-thaw, tidal, geothermal, or other cycling environments can alternately mix reactants, concentrate them, and then preserve a fraction of the resulting products [1, 2]. The claim is not that one setting is proven to be universal. The claim is that an origin sequence needs some mechanism that periodically changes effective water activity, concentration, retention, and degradation pressure so that polymer-length or pattern-memory gains can accumulate.

1 Introduction

Papers 2–4 established geochemical redox, electron transport, and proton/ion coupling as the first concrete substrate. That is not enough. A system also needs a route from small molecules to longer-lived chemical patterns. Polymerization is difficult in bulk water because condensation reactions compete with hydrolysis and product loss. CDFD therefore asks what environmental topology can raise the forward polymerization opportunity without permanently destroying transport. This includes wet-dry cycling, mineral-surface oligomerization, and activated nucleotide chemistry as different ways to change the same constraint landscape [6, 5].

2 Oscillating Adaptive Ratios

The shared adaptive ratio is

$$\Psi_s = \left(\frac{\Phi}{C} \right) S M_s. \quad (1)$$

In a cycling environment, the relevant quantities are time dependent:

$$\Psi_s(t) = \left(\frac{\Phi_{\text{chem}}(t)}{C_{\text{hydro}}(t)} \right) S(t) M_s(t). \quad (2)$$

Wet phases can increase encounter rates and transport. Dry or concentrated phases can raise monomer activity and reduce the effective hydrolytic penalty for selected reactions. Rehydration then tests whether any products survive long enough to seed another cycle.

3 The Polymerization Ratchet

A minimal water-volume cycle can be written

$$V_{\text{H}_2\text{O}}(t) = V_0 + \Delta V \cos(\omega t). \quad (3)$$

As $V_{\text{H}_2\text{O}}$ falls, monomer concentration rises and the system can enter a condensation-favoring window. As water returns, hydrolysis and dilution resume. A ratchet exists only if the retained product fraction after one cycle is greater than the lost fraction over many cycles:

$$M_s(t + nT) > M_s(t) \quad (4)$$

for at least some range of periods T , temperatures, solute mixtures, surfaces, and degradation rates.

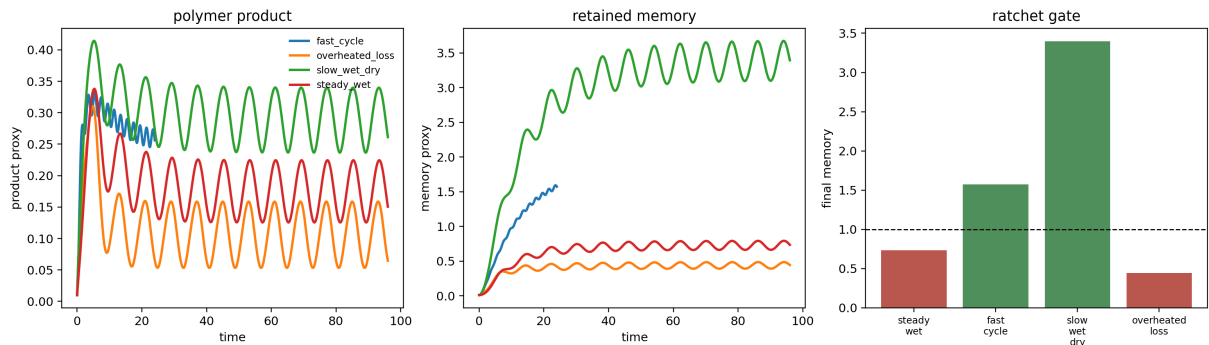


Figure 1: Wet-dry polymerization-ratchet diagnostic. The diagnostic separates product formation from retained memory across several cycling regimes. The model supports a conditional ratchet claim: activation, concentration, retention, and survival must align over repeated cycles.

4 Recent Experimental Anchors

The cycling argument has strengthened since older wet-dry sketches. Heat-flow crack experiments show that geological non-equilibria can concentrate and purify many prebiotically relevant molecules while enhancing reaction yields [3]. Related work on phosphate shows that heat flows can solubilize apatite-derived phosphate and support trimetaphosphate formation under conditions relevant to prebiotic activation [4]. Meanwhile, aqueous-origin reviews emphasize that condensation polymer growth is not solved by saying “remove water”; useful chemistry must store activation energy in precursors, release it selectively, and survive hydrolysis pressure [7]. In CDFD terms, the ratchet is not only wet versus dry. It is the repeated alignment of activation, concentration, retention, and survival.

Status: Inferred

5 Claim Boundary

The claim is not that static submerged vents are impossible or that hot springs are mandatory. The claim is that a complete CDFD origin chain must include an environmental or material mechanism that solves the same functional problem:

- periodic concentration or compartmental concentration,
- partial dehydration or equivalent activation chemistry,
- product retention against washout,
- survival of useful products across the next destructive phase.

Wet–dry cycling is one strong implementation of that function; mineral surfaces, eutectic phases, gels, and compartment interiors may supply related effects.

6 Bridge to Paper 6

The output of this step is not life. It is a population of longer-lived patterns, oligomers, and surface-bound products. Paper 6 asks when such patterns become self-reinforcing through autocatalytic closure and chemical memory.

7 Conclusion

Oscillating constraints provide a plausible way to turn environmental fluctuation into retained chemical structure. In the Part II chain, this paper supplies the transition from geochemical throughput to memory-bearing chemistry while keeping the historical setting open to empirical testing.

8 Reproducibility Note

The companion script `supplementary_ool_05.py` writes `outputs/paper_05/wet_dry_polymerization_ratche` plus CSV/JSON summaries. No notebook is required.

References

- [1] Bruce Damer and David Deamer. The hot spring hypothesis for an origin of life. *Astrobiology*, 20(4):429–452, 2020.
- [2] James P. Ferris, Aubrey R. Hill, Rihe Liu, and Leslie E. Orgel. Synthesis of long prebiotic oligomers on mineral surfaces. *Nature*, 381:59–61, 1996.
- [3] Thomas Matreux, Paula Aikkila, Bettina Scheu, Dieter Braun, and Christof B. Mast. Heat flows enrich prebiotic building blocks and enhance their reactivity. *Nature*, 628(8006):110–116, 2024.

- [4] Thomas Matreux, Almuth Schmid, Mechthild Rappold, et al. Heat flows solubilize apatite to boost phosphate availability for prebiotic chemistry. *Nature Communications*, 16:1809, 2025.
- [5] Bhavesh H. Patel, Claudia Percivalle, Dougal J. Ritson, Colm D. Duffy, and John D. Sutherland. Common origins of RNA, protein and lipid precursors in a cyanosulfidic protometabolism. *Nature Chemistry*, 7:301–307, 2015.
- [6] Matthew W. Powner, Béatrice Gerland, and John D. Sutherland. Synthesis of activated pyrimidine ribonucleotides in prebiotically plausible conditions. *Nature*, 459:239–242, 2009.
- [7] Daniel Whitaker and Matthew W. Powner. On the aqueous origins of the condensation polymers of life. *Nature Reviews Chemistry*, 8:817–832, 2024.